provide a new composition technique when functions are used as return values.

[4] SECD stands for Stack Environment Code Dump.

3.6.1. Global state problems

In listing 3.4 you saw the contents of a file stored in a variable after it was read, so that it could be used by another function. Suppose you have two files, tweedle.dum and tweedle.dee, and you always want to serve the contents of the one that was most recently modified. Suppose you want to implement this as a function that will serve the most recently modified file:

```
serveMostRecentFile 'tweedle.dum', 'tweedle.dee'
```

This will involve another piece of function glue—functions as return values.

3.6.2. Functions as return values

Suppose now you don't want to just serve the most recent file. Instead, you want to use the most recent file's contents somewhere else in your program. As it's written now, though, because you've contained the variables inside a scope, you're unable to access them in other parts of your program. You need a way to expose a specific variable from a scope to other parts of your program, without resorting to a global variable. To do this, you write another function. This function is going to use a new piece of programming glue that you've not seen yet, by using a *function as the return value from a function*. In order to get there and understand what's happening, it's worth first looking at a simple example of the concept to see how it works.

Abstract example

A simple, abstract example that illustrates the concept of returning a function from a function is a good place to start: